

UNIT DIGIT

① Level 1

$\{0, 1, 5, 6\}$

→ Cyclicity = 1

$$1^1 = 1$$

$$1^2 = 1$$

$$1^3 = 1$$

⋮

$$1^n = 1$$

$$5^1 = 5$$

$$5^2 = 2\underline{5} = 5$$

$$5^3 = 125 = 5$$

⋮

$$5^n = 5$$

$$6^1 = 6$$

$$6^2 = 3\underline{6} = 6$$

$$6^3 = 21\underline{6} = 6$$

⋮

$$6^n = 6$$

$$5^3 = 5^2 \times 5$$

$$= \cancel{2} \underline{5} \times \underline{5}$$

$$= 5 \times 5$$

$$= \cancel{5}$$

$$= 5 //$$

$$6^3 = 6^2 \times 6$$

$$= \cancel{3} \underline{6} \times \underline{6}$$

$$= \cancel{3} 6$$

$$= 6 //$$

Ex:

$$\textcircled{1} \quad (\cancel{2}\cancel{3}\cancel{1}\cancel{4}\cancel{7}\underline{1})^{1000} = \underline{1}^{1000} = \underline{1}$$

$$\textcircled{2} \quad (567\underline{1})^{10} = 1$$

$$\textcircled{5} \quad (555)^{555} = 5$$

$$\textcircled{3} \quad (1\underline{00})^{201} = 0$$

$$\textcircled{4} \quad (117\underline{66})^{21} = 6$$

② Level 2 $\rightarrow$ cyclicity 2 $\rightarrow$ [4, 9]

$$4^1 = \underline{4}$$
$$4^{\textcircled{2}} = 1\underline{6} = \textcircled{6}$$

$$4^3 = 6\underline{4} = \underline{4}$$

$$4^{\textcircled{4}} = 25\underline{6} = \textcircled{6}$$

$$y^{\text{odd}} = 4$$

$$y^{\text{even}} = 6$$

$$9^1 = 9$$

$$9^2 = 81 = 1 \checkmark$$

$$9^3 = \cancel{81} \times 9 = 9$$

$$9^4 = \cancel{81} \times \cancel{81} = 1 \times 1 = 1 \checkmark$$

$$9^{\text{odd}} = 9$$

$$9^{\text{even}} = 1$$

Ex :

$$\textcircled{1} \quad 5^{400} + 6^{600} = 5 + 6 = 11 = 1 //$$

$$\begin{aligned} \textcircled{2} \quad 4^{\overbrace{73}^{\circ}} + 4^{74} &= 4^0 + 4^e \\ &= 4 + 6 \\ &= 10 \\ &= 0 // \end{aligned}$$

$$\textcircled{3} \quad (\cancel{1}\cancel{1}9)^{999} = 9^0 = 9 //$$

③ Level 3 $\rightarrow$ cyclicity = 4 $\rightarrow$ [2, 3, 7, 8]

$$2^1 = 2 \checkmark$$

$$2^2 = 4 \checkmark$$

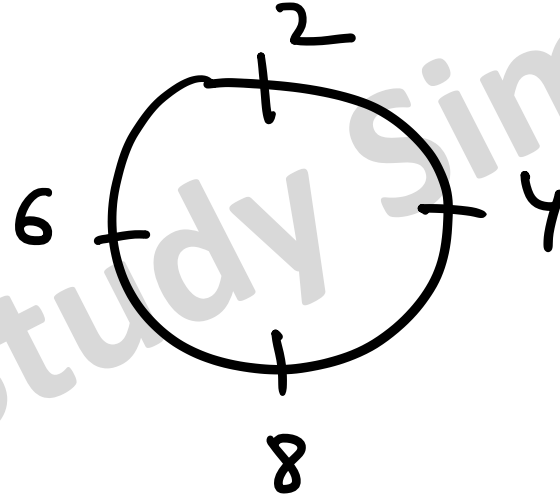
$$2^3 = 8 \checkmark$$

$$2^4 = 16 = 6 \checkmark$$

$$2^5 = 2$$

$$2^6 = 4$$

$$2^7 = 8, 2^8 = 6$$



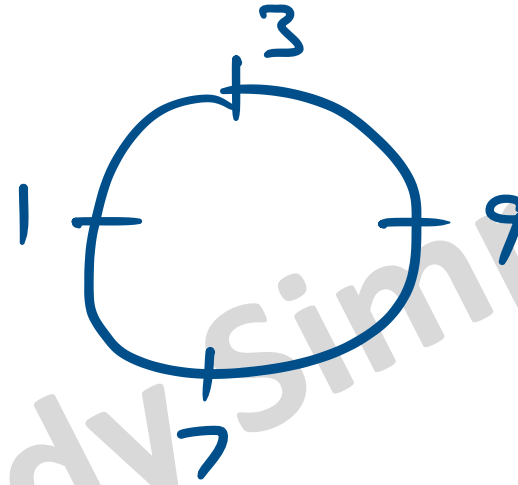
$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 9 \times 3 = 7$$

$$3^4 = 9 \times 7 = 1$$

$$3^5 = 3$$

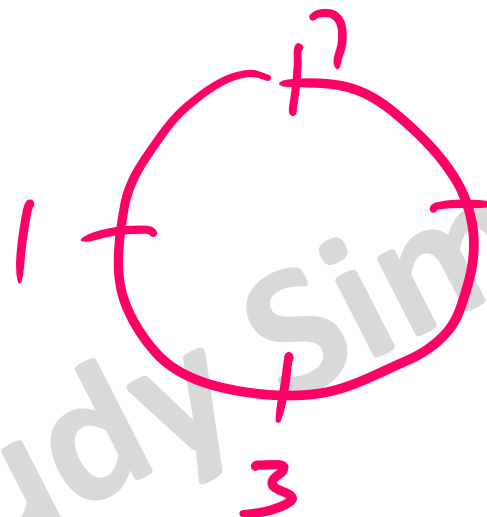


$$\therefore 7^1 = 7$$

$$7^2 = 49 = 9$$

$$7^3 = 343 = 3$$

$$7^4 = 49 \times 49 = 1$$

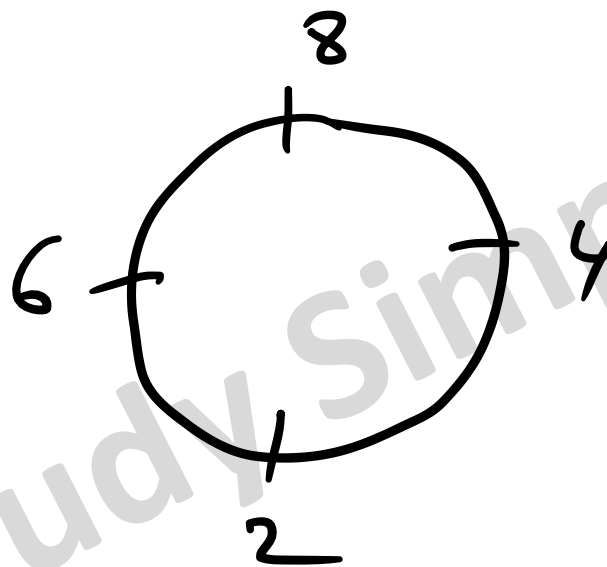


$$8^1 = 8$$

$$8^2 = 4$$

$$8^3 = 2$$

$$8^4 = 6$$



Ex : $2^{18} = 2^4 \cdot 2^4 \cdot 2^4 \cdot 2^4 \cdot 2^2$

$$= \underline{6 \cdot 6 \cdot 6 \cdot 6} \cdot 4$$

$$= (\cancel{6}) (\cancel{6}) (\cancel{6}) (4)$$

M1

$$= (\cancel{6}) 4$$

$$= \cancel{6} 4$$

$$= 4 //$$

Observation

$$2^{18} = \underline{2^4 \cdot 2^4 \cdot 2^4 \cdot 2^4} \cdot 2^2$$

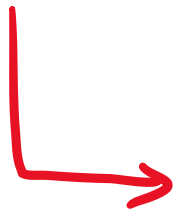
$$= \underline{2^{16}} \cdot 2^2$$

$$= \underline{(65536)} \cdot (4)$$

$$= 4$$

$$\checkmark \text{ Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

2 18



Cyclivity = 4

1) Cyclivity = 4
2)

$$\begin{array}{r} 4 \checkmark \\ 4 \overline{) 18} \\ \underline{-16} \\ 2 \checkmark \end{array}$$

$$18 = (\cancel{4 \times 4}) + 2$$

$$\therefore 2(\cancel{4 \times 4}) + 2 \leftarrow \text{remainder}$$

$$= \cancel{2^{16}} \cdot (2^2)$$

$$2^{18} = 2^2 = 4$$

$$\cdot \quad 2^n = 2^{(\Delta \times \Phi) + R} \checkmark$$

$$\Rightarrow 2^n = 2^R$$

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$$8] \quad 2^{\overline{77}}$$

$$(1) \quad C = 4$$

(2)

$$\begin{array}{r} 19 \\ 4 \overline{) 77} \\ \underline{- 4} \\ 37 \\ \underline{- 36} \\ 1 = r \end{array}$$

$$(3) \quad 2^{\overline{77}} = 2^r = 2^1 = 2_{//}$$

Q] 2^{1037}

① $c = 4$

②
$$\begin{array}{r} 9 \\ 4 \overline{) 37} \\ \underline{-36} \end{array}$$

$$\underline{1} = r$$

③ $2^{1037} = 2^r = 2^1 = 2 //$

Ex 1 : $(\cancel{4}7) 5398$

Soln : $7 \overset{5}{} \overset{3}{} \overset{9}{} \overset{8}{}$

① $c = 4$

②
$$\begin{array}{r} 24 \\ 4 \overline{) 98} \\ \underline{-8} \\ 18 \\ \underline{-16} \\ 2 = r \end{array}$$

③ $7 \overset{5398}{} = 7^r$
 $= 7^2$
 $= 9 //$

Ex 2 :

$$8^{\underline{1000}}$$

① $C = 4$

②
$$4 \overline{) 0000} \\ \underline{-0000} \\ 0000 = 2$$

③ $8^{10000} = 8^0 = 1$ $\bigotimes$
 $= 8^4 = 6 //$ $\checkmark$

Ex 3 : $\cancel{1}2^{12} \times \cancel{2}2^{22}$

Solⁿ : $2^{12} \times 2^{22}$

$= 2^{12+22}$

$= 2^{34}$

③ $2^{34} = 2^2 = 4$

① $C = 4$

②

$$\begin{array}{r} 8 \\ 4 \overline{) 34} \\ \underline{-32} \\ 2 \end{array}$$

Ex 4 : $\cancel{1} \cancel{3} 3^{133}$

solⁿ : 3^{133}

① $c = 4$

②
$$\begin{array}{r} 8 \\ 4 \overline{) 33} \\ \underline{-32} \\ 1 \end{array}$$

③ $3^{133} = 3^1 = 3 //$

Ex 5 : $133^{63} \times 4^{49}$

Soln : $3^{63} \times 4^{49}$

$$3^{63} \times 4^0$$

$$3^3 \times 4$$

$$7 \times 4$$

$$= 8 //$$

$$\begin{array}{r} 15 \\ 4 \overline{) 63} \\ \underline{-4} \\ 23 \\ \underline{-20} \\ 3 \end{array}$$

$$Q] \quad \underline{4}^{69} \times 6^5 = 4^0 \times 6^4 = 4 \times 6 = 4 //$$

$$Q] \quad 18^{18} \times 28^{28} \times 288^{288}$$

Soln : $8^{18} \times 8^{28} \times 8^{288}$

$$8^{334}$$

$$8^2 = 64 = 4 //$$

$$\begin{array}{r} 8 \\ 4 \overline{) 34} \\ \underline{-32} \\ 2 \end{array}$$

Q] $11'' + 12^{12} + 13^{13} + 14^{14} + 15^{15}$

Soln

$$1'' + 2^{12} + 3^{13} + 4^{14} + 5^{15}$$

$$1 + 2^4 + 3^1 + 6 + 5 = 1 + 6 + 3 + 1$$

$$= 10 + 1$$

$$= 11$$

$$= 11$$

$$\begin{array}{r} 3 \\ 4 \overline{) 12} \\ \underline{-12} \\ 0 \end{array}$$

$$\begin{array}{r} 3 \\ 4 \overline{) 13} \\ \underline{-12} \\ 1 \end{array}$$

Q] Unit position of

$$\cancel{21}^{870} + \cancel{146}^{127} \times 3^{424}$$

Soln:

$$1^{870} + (6^{127} \times 3^{424})$$

$$1 + (6 \times 3^4)$$

$$1 + (6 \times 1)$$

$$1 + 6$$

$$= 7 //$$

$$\begin{array}{r} 6 \\ 4 \overline{) 24} \\ \underline{-24} \\ 0 \end{array}$$

$$0! = 1$$

$$1! = 1$$

$$2! = 2$$

$$3! = 6$$

$$4! = 24$$

$$5! = 120$$

...

(÷ by 4)

$$\# n! = n(n-1)!$$

$$1! = 1(1-1)!$$

$$1! = 1(0)!$$

$$0! = 1! = 1$$

• 9] $2^{11!} = 2^0 = 2^4 = 6 //$

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If $(1.001)^{1259} = 3.52$ and

$(1.001)^{2062} = 7.85$ then $(1.001)^{3321} =$

[EC, EE, IN, MT]

~~(A) 2.23~~

~~(B) 4.33~~

~~(C) 11.37~~

~~(D) 27.64~~

$$\checkmark a^{1259} = 3.52 - 1$$

$$\checkmark a^{2062} = 7.85 - 2$$

$$a^{3321} = (3.52)(7.85)$$

$$a^m \cdot a^n = a^{m+n}$$

Find the sum to n terms of the series $10 + 84 + 734 + \dots$ [EC, EE, IN]

☒ (A) $\frac{9(9^n + 1)}{10} + 1$

☒ (B) $\frac{9(9^n - 1)}{8} + 1$

☒ (C) $\frac{9(9^n - 1)}{8} + n$

☒ (D) $\frac{9(9^n - 1)}{8} + n^2$

MI

$n=1$: $S = 10$

A, B, C, D

$n=2$: $S = 94$

Q2

$$S = 10 + 84 + 734 + \dots$$

$$= (9+1) + (81+3) + (729+5) + \dots$$

$$= (9 + 9^2 + 9^3 + \dots) + (1 + 3 + 5 + \dots)$$

GP for n terms

AP for n terms

$$= n^2$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{9(9^n - 1)}{8}$$

$$1 + 3 + 5 + \dots$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} [\cancel{2}(1) + (n-1)\cancel{2}]$$

$$= \cancel{\frac{n}{2}} (\cancel{2}n)$$

$$= n^2 //$$

Find the sum of the expression

$$\frac{1}{\sqrt{1+\sqrt{2}}} + \frac{1}{\sqrt{2+\sqrt{3}}} + \frac{1}{\sqrt{3+\sqrt{4}}} + \dots + \frac{1}{\sqrt{80+\sqrt{81}}}$$

[CS, ME, PI]

(A) 7

✓ (B) 8

(C) 9

(D) 10

$$= \frac{\sqrt{1}-\sqrt{2}}{(1-2)} + \frac{\sqrt{2}-\sqrt{3}}{(2-3)} + \frac{\sqrt{3}-\sqrt{4}}{(3-4)} + \dots + \frac{\sqrt{80}-\sqrt{81}}{(80-81)}$$

$$= -1 \left[\sqrt{1} - \cancel{\sqrt{2}} + \cancel{\sqrt{2}} - \cancel{\sqrt{3}} + \cancel{\sqrt{3}} - \cancel{\sqrt{4}} + \dots - \cancel{\sqrt{80}} + \sqrt{81} \right]$$

$$\cdot = \sqrt{81} - \sqrt{1}$$

$$= 9 - 1$$

$$= 8 //$$

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Fill in the missing number in the series.

2 3 6 15 157.5 630
↑ ↑ ↑ ↑ ↑
[EC-2, ME-2]

$$\frac{3}{2} = 1.5$$

$$\frac{6}{3} = 2$$

$$\frac{15}{6} = 2.5$$

$$\frac{x}{15} = 3$$

$$\rightarrow x = 3 \times 15 = 45 //$$

What is the next number in the series?

12 35 81 173 357 $\overline{x}$
[EC-1, ME-1]

23 46 92 184 368
 $\swarrow$ $\swarrow$ $\swarrow$ $\swarrow$
 $\times 2$ $\times 2$ $\times 2$ $\times 2$

$$x - 357 = 368$$

$$x = 368$$

$$\begin{array}{r} + 357 \\ \hline 725 \end{array}$$

If $\left(z + \frac{1}{z}\right)^2 = 98$, compute $\left(z^2 + \frac{1}{z^2}\right)$.

[CS-1, EE-1]

$$\textcircled{z^2} + 2z \cdot \frac{1}{z} + \textcircled{\frac{1}{z^2}} = 98$$

$$z^2 + \frac{1}{z^2} = 98 - 2 = 96 //$$

The value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$ is

[CS-2, EE-2]

(A) 3.464

(B) 3.932

~~(C) - 3~~

☒ (C) 4.000

(D) 4.444

$$y = + \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$$

$$y = \sqrt{12 + y}$$

$$y^2 = 12 + y \Rightarrow y^2 - y - 12 = 0$$

$$(y - 4)(y + 3) = 0 \Rightarrow y = 4, -3$$

Which number does not belong in the series below? [CS-3, EE-3]

2, 5, 10, 17, 26, 37, 50, 64

(A) 17

(B) 37

☒ (C) 64

(D) 26

3, 5, 7, 9, 11, 13, 14

The missing number in the given sequence 343 , 1331 , $\frac{2197}{13^3}$, 4913 is

(a) 4096

~~(b) 2744~~

(d) 2197

~~(c) 3375~~

$$\begin{array}{r} \checkmark \\ 11 \times 11 \\ \hline \end{array} = 121$$

$$\begin{array}{r} \checkmark \\ 121 \times 11 \\ \hline \end{array} = 1331$$

Find the smallest number y such that $y \times 162$ is a perfect cube. [CS-1, EE-1]

(A) 24

(B) 27

(C) 32

~~(D) 36~~

$$y \times 162 = \text{perfect cube}$$

$$y \times 2 \times 9 \times 9$$

$$y \times 2 \times 3 \times 3$$

$$2^2 \times 3^2$$

$$\begin{aligned} \text{Ans} &= 4 \times 9 \\ &= 36 // \end{aligned}$$

$$\begin{array}{r|l} 2 & 162 \\ \hline & 81 \\ 9 & 9 \\ \hline & 1 \end{array}$$

The sum of the digits of two digits number is 12. If the new number formed by reversing the digit is greater than the original number by 54, find the original number.

(A) 39 (B) 57 (C) 66 (D) 93 [CE 2016]

Logic :

$$\begin{array}{l} \underline{25} = \underline{2 \times 10 + 5} \\ \underline{xy} = \underline{10 \times x + y} \end{array}$$

$$yx = 10 \times y + x$$

(M1)

① $x + y = 12$

② $(10 \times y + x) - (10 \times x + y) = 54$

$$\Rightarrow x - y = -6$$

$$\begin{array}{l} x = 3 \\ y = 9 \end{array}$$

New no $>$ 09 by 54

New no = 09 + 54

New no - 09 = 54

The sum of the digits of two digits number is 12. If the new number formed by reversing the digit is greater than the original number by 54, find the original number.

(A) 39 (B) 57 (C) 66 (D) 93 [CE 2016]

$$(93) - (39) = 54$$

~~(A) 3 + 9~~

(B) 5 + 7

(C) 6 + 6

(D) 9 + 3

$$93 - 39 = 54$$

$$75 - 57 = 18$$

$$66 - 66 = 0$$

$$39 - 93 = -ve$$

m2